Oral script: the 10-minute defence

Museum Guide project, MSA-DATB09-03. One section per slide, to read aloud.

How to use it

- About 1260 words: at a calm pace (about 130 words per minute), this is about 10 minutes. Without the 83 optional words, about 9 minutes. The rest of the 10 minutes is a safety margin for pauses, clicks and nerves.
- Every slide ends with a green TRANSITION box, always built the same way: "We saw that... So now...". Say it while you click to the next slide, and never skip it, even when you are late: it turns the six steps into one story.
- Read it aloud 3 times, with the slides, and time yourself once. Speak slowly; pause after each bold sentence: it is the one the jury must remember.
- Blue text in brackets is a stage cue (where to point); do not read it. Grey sentences marked (optional) can be skipped if you are late.
- If you present with a teammate, a natural split is slides 1 to 6, then slides 7 to 11.

The story, transition by transition

1 to 2: To answer it, we did not jump straight to a model. So now, here is our path: six steps.

2 to 3: So now, let's start with step one: the data, and how we made the test fair.

3 to 4: Our data is ready and our test is fair. So now, step two: our first model.

4 to 5: We saw that our MLP is overfitting. So now, step three: we try to fix this overfitting.

5 to 6: We saw that the overfitting is under control. But our MLP still sees colours, not shapes. So now, step four: we need a network that sees shapes.

6 to 7: We saw how ResNet18 works. So now, let's see what it gives on our paintings.

7 to 8: But to answer the director, we also need a challenger without any neural network. So now, step five: our classical challenger, then the final test.

8 to 9: We saw how our challenger works. So now, the final tests: every model, on paintings never seen before.

9 to 10: Accuracy is only half of the director's question. So now, step six: what does it cost on a kiosk?

10 to 11: We saw that it is accurate, fast and cheap. So now, we have everything we need to answer the director.

Slide 1 · Is a neural network worth it on a museum kiosk?

0:00 to 0:40 · 40 s · 84 words

Good morning. Our museum wants kiosks that recognise the style of a painting from a photo.

We worked on five styles: Baroque, Cubism, Impressionism, Pop Art and Ukiyo-e. *[point at the five paintings]*

The kiosks have small CPUs, the team is small and the budget is limited. So the director asks one question: **is a neural network worth it in production, or would a simpler classical model do the job?**

> **TRANSITION TO SLIDE 2**
> To answer it, we did not jump straight to a model. **So now, here is our path: six steps.**

Slide 2 · Six steps to answer the director

0:40 to 1:10 · 30 s · 68 words

[point at the six steps, one by one] Step one: the data, and a fair test. Step two: our own neural network. Step three: we monitor its training and fix it. Step four: a pretrained ResNet18. Step five: the final tests, against a classical model. Step six: the cost on a kiosk.

Each step told us what to try next.

> **TRANSITION TO SLIDE 3**
> **So now, let's start with step one: the data, and how we made the test fair.**

Slide 3 · A fair test: one painter, one part

1:10 to 1:55 · 45 s · 101 words

We took 1 500 paintings from WikiArt, 300 per style.

We cut them into three piles: 1 061 to train, 219 to validate, that is, to choose our settings, and 220 to test, only once, at the end.

One rule matters a lot: **all the paintings of a known painter go to the same pile.** *[point at the zero]* With a random split, 23 of our 32 known painters would be in two piles, and a model could learn to recognise Monet instead of Impressionism. With our split: zero.

> **TRANSITION TO SLIDE 4**
> Our data is ready and our test is fair. **So now, step two: our first model.**

Slide 4 · Our MLP learns the paintings by heart

1:55 to 2:50 · 55 s · 116 words

Our first model is the simplest neural network of the course: a multi-layer perceptron built from nn.Linear layers and ReLU.

Before trusting it, we ran two checks. First, a model that knows nothing should have a loss of ln 5, about 1.61. *(optional) Ours started at 1.99, too confident, so we made its last layer smaller and got 1.59.* Second, it must be able to learn one small batch by heart: it does, 100 percent.

[point at the curves] The training accuracy goes to 99.7 percent, but the validation accuracy stays around 40 percent. **This is overfitting: the network learns the paintings by heart.**

> **TRANSITION TO SLIDE 5**
> We saw that our MLP is overfitting. **So now, step three: we try to fix this overfitting.**

Slide 5 · Healthy training, but the MLP overfits

2:50 to 3:45 · 55 s · 120 words

Before fixing it, we looked inside the network to check that the training itself was healthy: the gradient reaches every layer and only 4 percent of the neurons are dead. So the problem really is overfitting.

Then we attacked it with the tools of session 6. *[point at the chart]* Together, weight decay, dropout, batch norm, flips and early stopping take the MLP from 38.8 to 45.5 percent, and the gap from 0.61 to 0.12.

But one number shows a deeper limit: **the MLP finds only 7 percent of the Cubist paintings. It sees colours, not shapes.**

> **TRANSITION TO SLIDE 6**
> We saw that the overfitting is under control. But our MLP still sees colours, not shapes.
> **So now, step four: we need a network that sees shapes.**

Slide 6 · ResNet18: a network made for images

3:45 to 5:25 · 100 s · 218 words

To see shapes, we need a network made for images: a convolutional network, or CNN. The brief asks for a ResNet, a famous family of CNNs from Microsoft Research, which won the big ImageNet competition in 2015. **ResNet18 is its smallest member, with 18 layers: light enough for a kiosk, and PyTorch gives it already trained.** *(optional) "Res" means residual: shortcuts between its layers let a deep network learn well.*

[point at the dark box] Its 18 layers slide small filters over the picture: the first layers find edges and colours, the next ones textures, the last ones shapes. *(optional) Our MLP saw 12 288 separate pixels.*

ResNet18 already learned this on 1.2 million ImageNet photos, and we only have 1 061 paintings, so we never retrain all of it. *[point at the 512 numbers]* It turns each painting into 512 numbers. *[point at the blue box]* We replace its last layer, made for ImageNet, by our own: from these 512 numbers to our 5 styles.

[point at the bottom right card] We tried two ways. **Frozen: only our new layer learns, 2 565 weights. Fine-tuned: the first layers stay frozen, but the last block learns again on our paintings, 8.4 million weights.** *(optional) Like hiring an experienced art expert: frozen, we only teach them our five style names; fine-tuned, we also retrain their eye for shapes.*

> **TRANSITION TO SLIDE 7**
> We saw how ResNet18 works. **So now, let's see what it gives on our paintings.**

Slide 7 · A pretrained ResNet18 sees shapes

5:25 to 6:15 · 50 s · 105 words

[point at the blue bar] The frozen version trains in less than a second and gives **88.1 percent on validation, almost twice our best MLP.** *[point at the top card]* And it does see shapes: it finds 71 percent of the Cubist paintings, against 7 for our MLP.

The fine-tuned version looked less convincing on validation, 85.4 percent, so we first chose the frozen one. **Keep this in mind.**

[point at the bottom card] We monitored it like the MLP: the gap between train and validation is only 0.09, against 0.61 for our first MLP.

> **TRANSITION TO SLIDE 8**
> But to answer the director, we also need a challenger without any neural network. **So now, step five: our classical challenger, then the final test.**

Slide 8 · Our classical challenger: a random forest

6:15 to 7:05 · 50 s · 111 words

A classical model cannot read raw pixels well. So we first describe each painting with 102 hand-made numbers: *[point at the second box]* 82 for its colours, and 20 for the direction and strength of its brush strokes.

[point at the tree] Then a random forest: 500 decision trees. Each tree asks yes or no questions on these numbers, like "many dark colours?", until it names a style. Each one learns from a random draw of the paintings, so they make different mistakes. **They vote, and the vote cancels the mistakes.**

[point at the right card] *(optional) Like every model, we chose its settings on validation: 63.5 percent.*

> **TRANSITION TO SLIDE 9**
> We saw how our challenger works. **So now, the final tests: every model, on paintings never seen before.**

Slide 9 · Final tests: the fine-tuned ResNet18 wins

7:05 to 8:05 · 60 s · 128 words

We opened the 220 test paintings only once, for every model. *[point at the grey bars]* The pretrained models are at the top, the random forest reaches 67, our MLP 41. But here is the surprise: the fine-tuned model was first, while it was second on validation.

Luck? We drew 220 new paintings from parts of WikiArt we had never opened, by painters our models never saw, and tested again. *[point at the blue bars]* Same ranking. *[point at the cards]* On the 440 paintings, even the frozen model is 18 points above the random forest, and fine-tuning wins by **5 points, between 2 and 8: it is not luck.**

(optional) Our 219 validation paintings were simply too few to separate the two versions.

> **TRANSITION TO SLIDE 10**
> Accuracy is only half of the director's question. **So now, step six: what does it cost on a kiosk?**

Slide 10 · Fast and cheap enough for a kiosk

8:05 to 8:45 · 40 s · 86 words

We timed one photo at a time, from the image to the answer. *[point at the blue card]* The fine-tuned ResNet needs **53 milliseconds on one CPU thread**, 27 with four threads: faster than a blink. Even on a kiosk four times slower, it answers in a fifth of a second.

The model file is 45 megabytes, and training takes two and a half minutes on a laptop, without a GPU.

> **TRANSITION TO SLIDE 11**
> We saw that it is accurate, fast and cheap. **So now, we have everything we need to answer the director.**

Slide 11 · Our choice: a fine-tuned ResNet18

8:45 to 9:40 · 55 s · 123 words

[point at the line under the title] On the 440 test paintings, our own MLP reached 42 percent, the random forest 65, and the fine-tuned ResNet18 88.

So, our answer to the director: yes, a neural network is worth it, but a pretrained one that we reuse. **We recommend ResNet18, pretrained on ImageNet, with its first layers frozen and its last block fine-tuned on our paintings.**

[point at the left column] Why? It is accurate: one visitor in eight gets a wrong style, against one in three. It is fast enough for a modest CPU, and cheap for a small team: retrained in a few minutes.

[point at the right column] Before the launch, we must test real photos taken at the kiosks, add more styles, and measure the real hardware.

Thank you. We are happy to take your questions.